

POC ARCHITECTURE REVIEW

The DevOps OverSight Agent

AI-assisted incident response that correlates **Splunk** and **Datadog** over MCP to diagnose and remediate incidents in a microservice mesh — two reference architectures.

Solution 1 · Ballerina + MCP Proxy

Solution 2 · LangChain + A2A

Agenda

01

The problem

Why cross-platform incident diagnosis is slow, risky, and manual today.

02

Level 0 architecture

The concepts both solutions share: the mesh, telemetry fan-out, MCP, correlation, the human-approval gate.

03

Solution 1 — Ballerina + MCP Proxy

One agent, one MCP endpoint that federates Splunk & Datadog behind lazy-loaded tools.

04

Solution 2 — LangChain + A2A

An orchestrator that delegates to Datadog & Splunk specialist agents over the A2A protocol.

05

Comparison & path to enterprise

An honest scorecard, the 5-minute demo, and what production hardening looks like.

Executive summary

- **The pain:** in a P1, engineers burn the first 20–40 minutes hand-correlating Splunk logs against Datadog metrics — under pressure, by hand.
- **What we built:** an AI agent that runs that cross-platform investigation automatically, then proposes a specific fix and waits for a human to approve it.
- **Two architectures:** same POC, same 5-minute demo, built two ways to show the trade-off — a Ballerina MCP Proxy and a LangChain A2A multi-agent system.
- **The durable bets:** MCP as the contract boundary, correlation kept in one reasoning context, and propose-before-act as a hard rule — never autonomous.

≤ 5 min

inject → diagnose → approve → recover

< 90s

AI diagnosis (cloud model)

0

vendor creds to demo

♦ **Bottom line**

Both approaches work. This deck gives you the scorecard to choose.

01



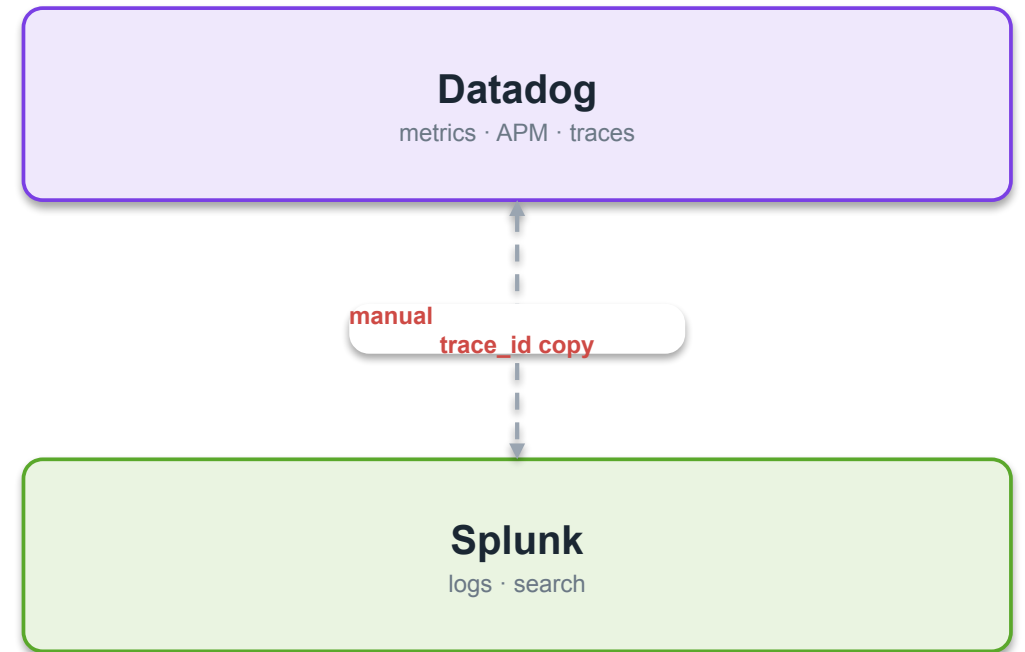
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The Problem

Why cross-platform incident diagnosis is slow, risky, and manual — and what 'good' looks like.

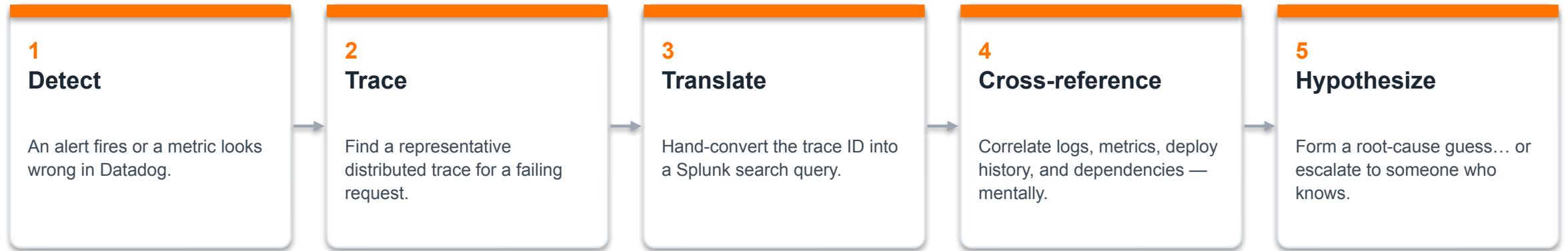
Best-of-breed tools, siloed by design

- **Two systems of record.** Logs and search live in Splunk. Metrics, APM, and traces live in Datadog. Both are excellent — and they don't share a query interface.
- **The join is manual.** During an incident an engineer swivel-chairs between them: spot a metric anomaly, pull a trace, copy the trace ID into a log search, cross-reference in their head.
- **It doesn't scale.** The work depends on individual institutional knowledge and gets harder as the mesh grows. Under pressure, at 2am, it's slow and error-prone.
- **Financial services raises the bar.** Any automation must be auditable, bounded, and must never change production without explicit human approval.



The engineer is the integration.

The manual-correlation tax



20–40 min
lost per major incident

♦ **Why this is the target**

Every one of these five steps is exactly what an LLM agent with the right tools can do — fast, consistently, and with its work shown.

Three forces shaped the design

Silo tax

Cut mean-time-to-diagnosis

Federate the two platforms so one investigation spans both — no context-switching, no manual trace-ID translation.

Approval gap

Middle path: assisted, not autonomous

Runbooks are auto-executed only after an explicit human approval. Enforced by the system, not by process or good intentions.

Lock-in risk

No AI vendor dependency

Model-agnostic by design — swap between a local model and cloud providers by config. Runs fully offline if needed.

Who we're building for

On-call / SRE Primary user in a live incident	Plain-language root cause + blast radius + a specific action to approve — with evidence links back to both platforms.
Platform / Observability Eng Operates & extends the system	Swap model or backend by config alone; the agent's own behavior is visible in the same stack it monitors.
Security & Compliance Governance, change mgmt, audit	No state change without prior approval; a fixed action allowlist; every action recorded; secrets never in code.
Demo / Sales Engineering Presents the capability	Full incident in ~5 minutes, offline, no vendor accounts, resettable to a clean state.
Service / App owners Own individual services	Ownership, runbooks, and SLA metadata surfaced per service so response routes correctly.

What 'good' looks like — the POC bar

< 90s

AI diagnosis (cloud)
Under 3 min on a local model

≤ 5 min

Full incident cycle
Inject → diagnose → approve → recover

2

platforms, every time
Evidence from BOTH Splunk & Datadog

0

vendor credentials
Full cycle runs offline

♦ Propose-before-act

The agent must list available runbooks and receive an approval signal before it ever executes one. No exceptions.

♦ Model portability

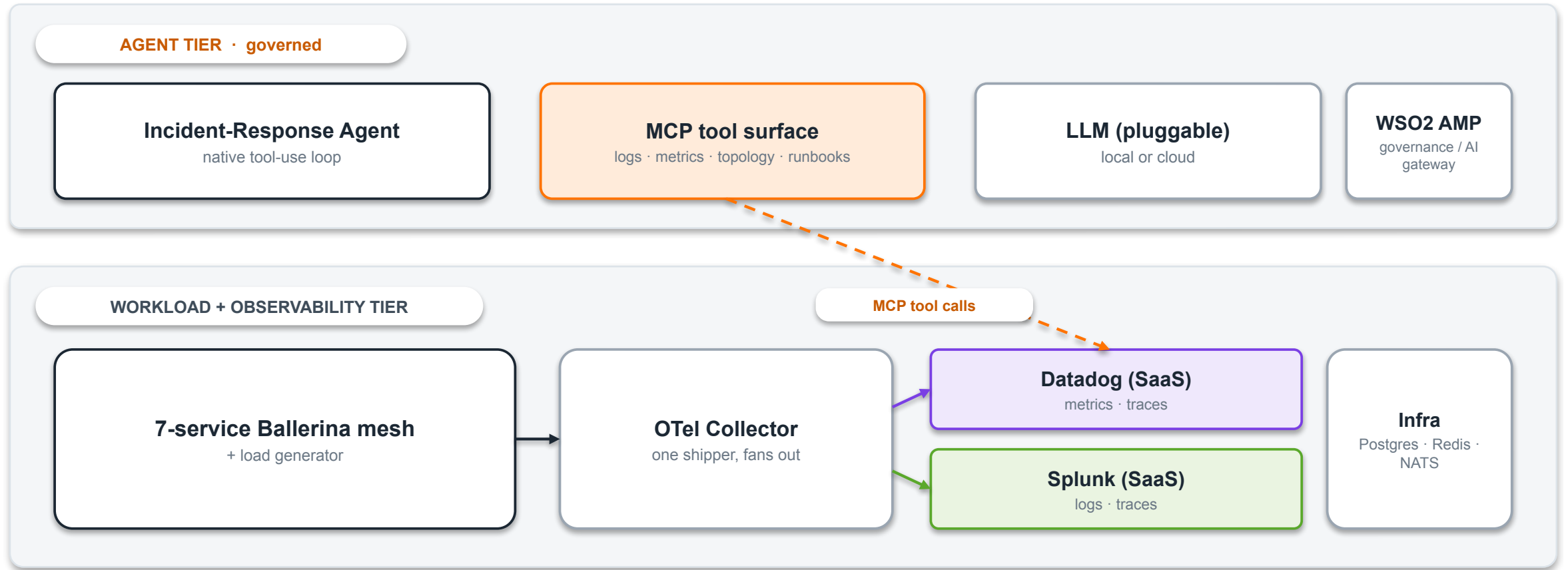
The same investigation must run on at least two providers (e.g. local Ollama and Anthropic) with no change beyond the provider selector.

SECTION

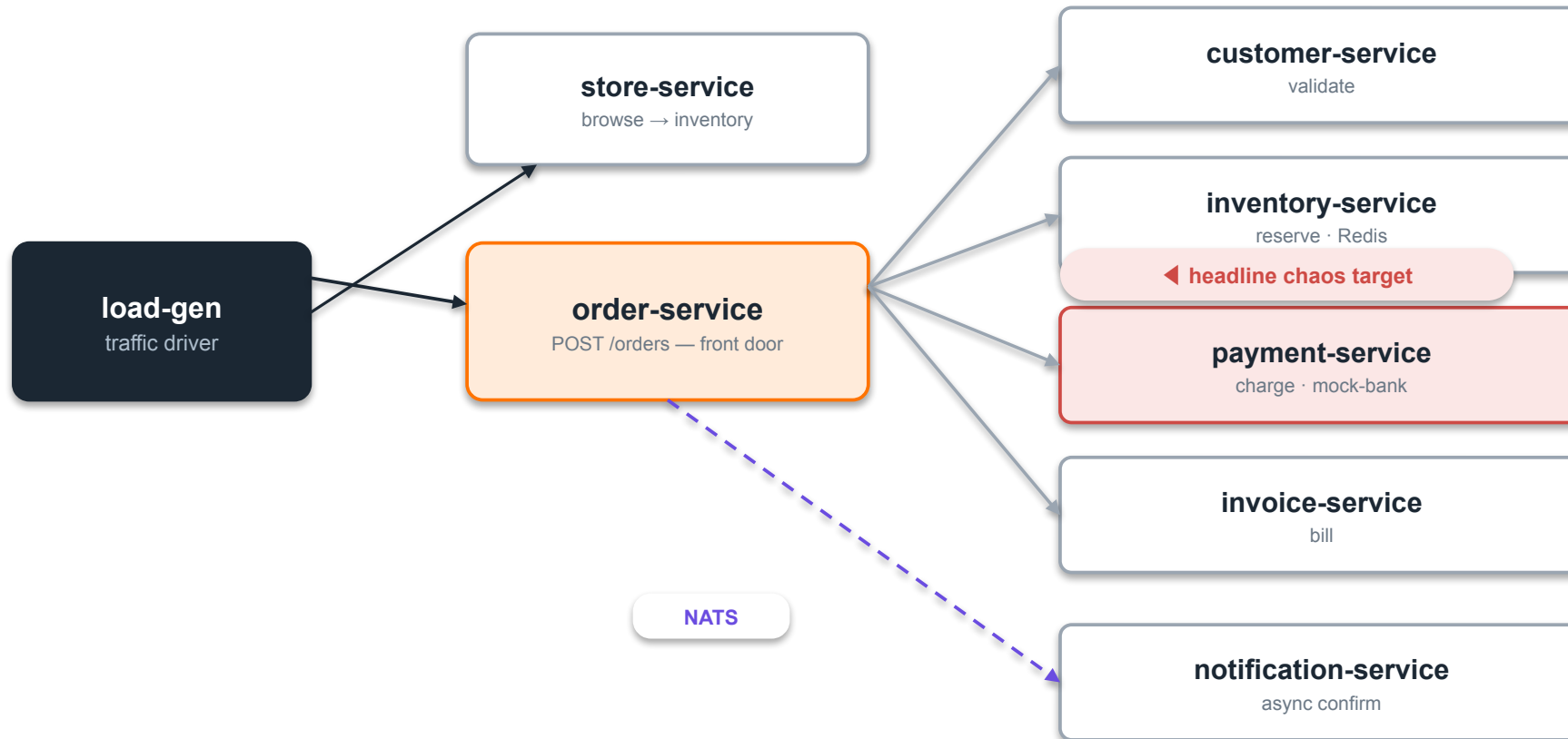
Level 0 Architecture

The concepts both solutions share: the mesh, telemetry fan-out, MCP, correlation, and the approval gate.

The big picture: two tiers



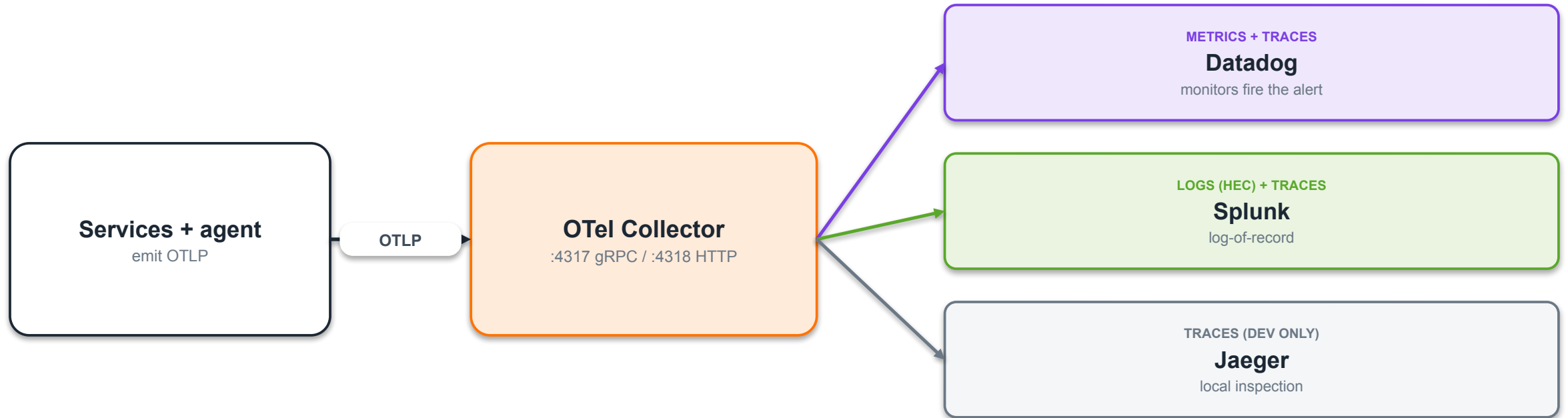
The mesh we observe & remediate



Why a mesh?

- Blast radius only exists across services.
- order fans out sync to 4 services + one async NATS hop to notification.
- Each service has a token-gated /chaos endpoint to inject faults.
- payment-service is the demo's failure point.

Telemetry fan-out: one collector, routed by signal

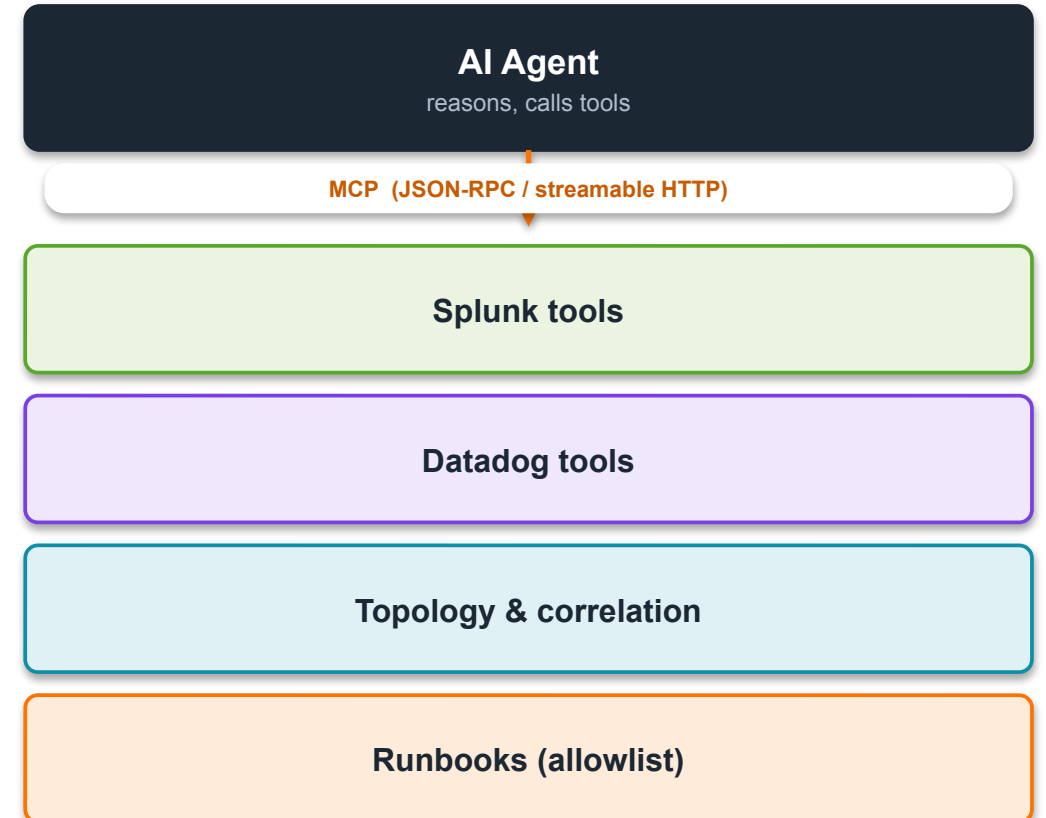


♦ The routing rule

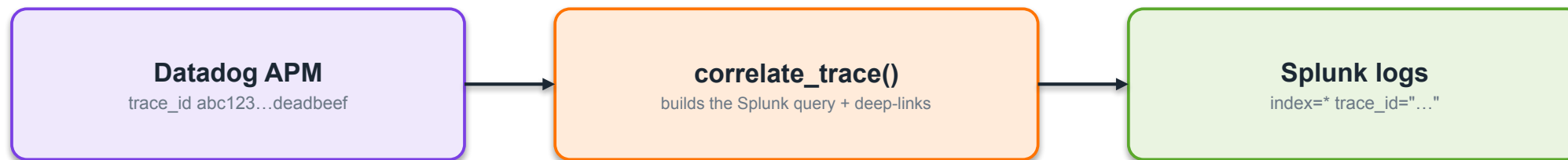
Traces → both platforms (so the agent can pivot by trace_id). Logs → Splunk only. Metrics → Datadog only. DD_LOGS_ENABLED=false avoids double-billing.

MCP: the contract boundary

- **Everything is a tool.** Splunk search, Datadog metrics, service topology, correlation, runbooks — the agent sees one uniform tool surface over MCP (Model Context Protocol).
- **Mock $\rightleftharpoons$ live is config, not code.** Swapping a mock backend for a real vendor MCP is a URL/token change. The agent's code never changes.
- **MCP fills what vendors can't.** Splunk MCP knows logs. Datadog MCP knows metrics. Neither knows YOUR catalog, dependency graph, owners, or runbooks — we add those as local tools.
- **One place for policy.** Routing, auth, and result hygiene get a single home at the boundary.



The core trick: correlate by trace_id



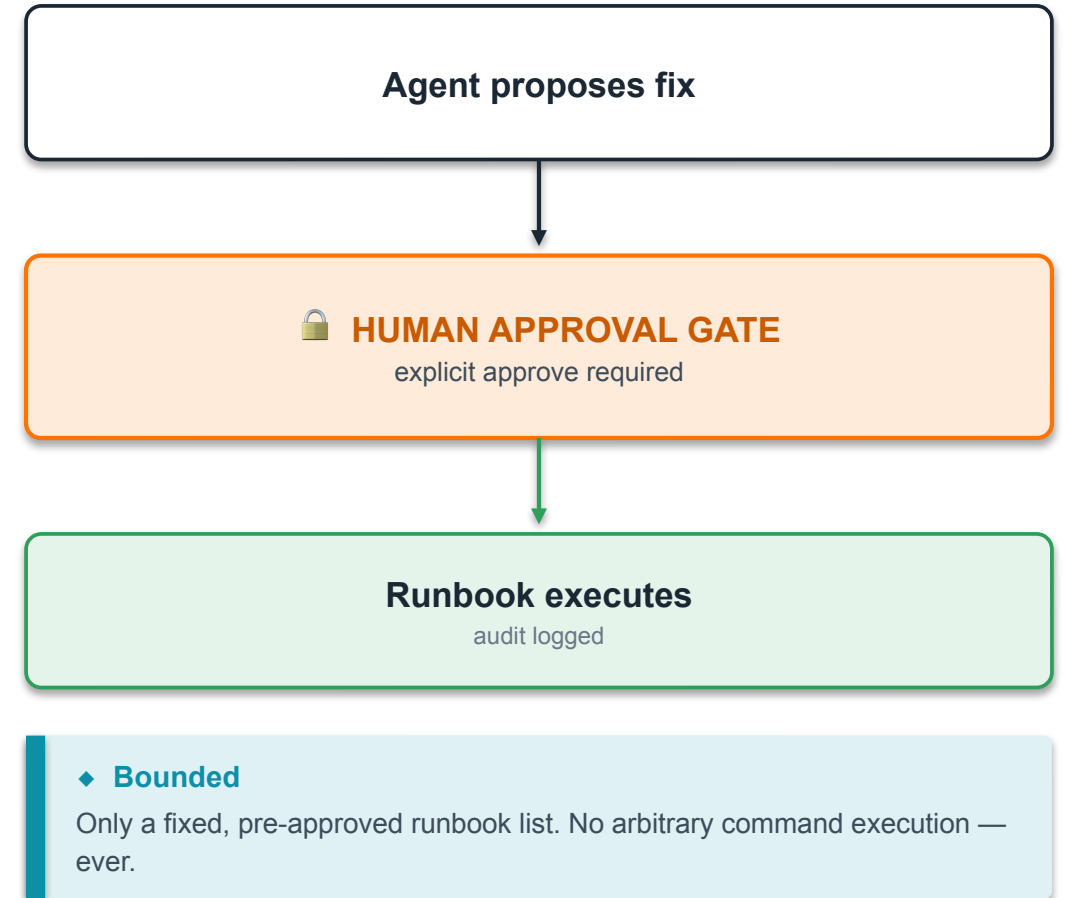
Both platforms carry the same trace_id in structured logs — that shared key is the whole game.

◆ ⚠ The #1 correctness gotcha — trace-ID width

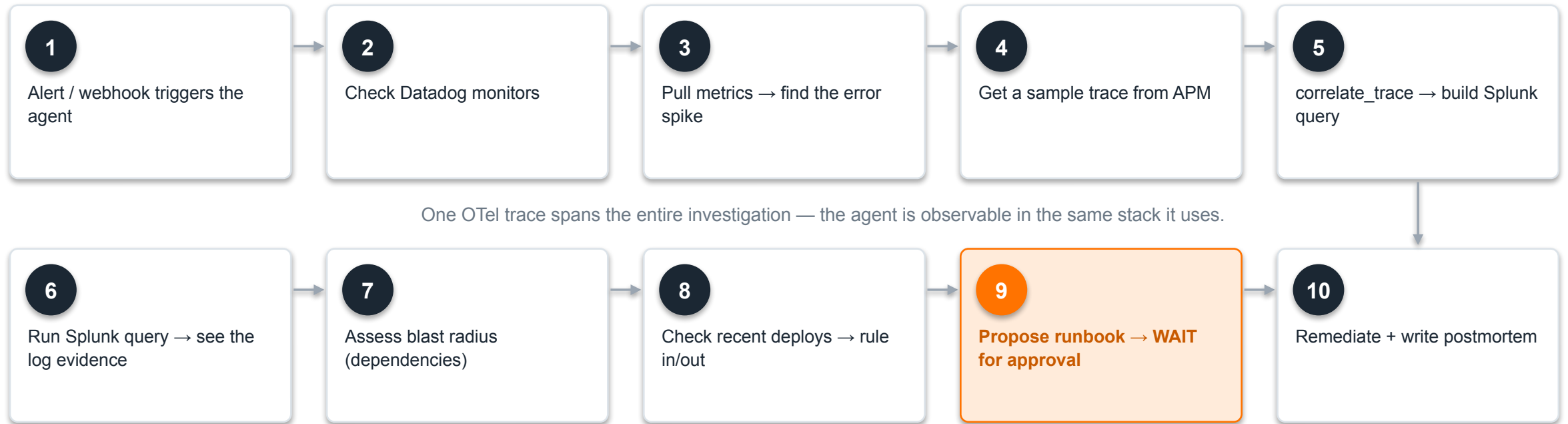
Datadog surfaces a 64-bit dd.trace_id; OpenTelemetry & Splunk hold a 128-bit trace_id. Search Splunk with the wrong width and the agent wrongly concludes "no logs found." The correlation layer MUST normalize both. (The mock's single demo ID masks this — real traffic exposes it.)

The agent's job — and the gate

- **Investigate.** On an incident signal, run a structured cross-platform investigation in a native tool-use loop.
- **Synthesize.** Produce a root-cause summary with blast radius and evidence links into both Splunk and Datadog.
- **Propose.** Name a specific, vetted runbook and the parameters it would use — e.g. disable-chaos on payment-service.
- **Wait.** Stop. Do nothing to production until a human approves. This is the hard line.
- **Remediate & record.** On approval, run the runbook, stream progress, write an audit entry, and post a postmortem.



End-to-end: the 10-step investigation



WSO2 Agent Manager's role

◆ Runtime & governance

Hosts the agent as a platform-managed workload from a Git project. Lifecycle, identity, and policy live here.

◆ AI gateway

All LLM traffic can route through AMP for rate-limiting, quota, model routing, and model-level audit — injected by config.

◆ Self-observability

The agent's own spans — reasoning, tool calls, token use, latency — flow through the same collector. Detect runaway agents.

◆ Honest scoping for tonight

In Solution 1 (Ballerina) AMP is the agent runtime. In Solution 2 (LangChain) the agents run as plain processes and AMP appears only as an optional LLM gateway. AMP is the enterprise governance plane — not a hard dependency of the POC.

No lock-in: the model is a config flag

Ollama LOCAL · CREDITS-FREE qwen2.5 / qwen3.5 family On-prem, zero cost, offline demo	Anthropic CLOUD claude-sonnet-4-6 Fastest, highest-quality tool use	OpenAI CLOUD gpt-4o Alternative managed provider	WSO2 AMP ENTERPRISE GATEWAY OpenAI-compatible Governed model access + audit
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Switch with one variable: **LLM_PROVIDER = ollama | anthropic | openai | amp**

The default demo runs fully offline on a local model — no vendor account required.

THE FORK

Two roads to the same goal

Everything so far is shared. Here's where the two solutions diverge — and it comes down to one question:

Where do the topology/correlation tools live, and how do we keep the agent's context small?

SOLUTION 1

Ballerina + MCP Proxy

One agent talks to one MCP Proxy that federates the backends and lazy-loads their tools. Low-context by a server-side registry.

SOLUTION 2

LangChain + A2A

An orchestrator delegates to specialist agents over A2A. Low-context by decomposition — each specialist owns only its own small toolset.

SECTION

Ballerina + MCP Proxy

One agent, one MCP endpoint that federates Splunk & Datadog behind lazy-loaded tools.

The core idea

- **One agent, one endpoint.** The Ballerina agent connects to exactly one MCP server — the MCP Proxy on :8290. It never touches Splunk or Datadog directly.
- **The proxy federates.** Behind that one endpoint it fans out to the Splunk and Datadog MCP backends and adds local topology, correlation, and runbook tools.
- **Small context by lazy loading.** The proxy hides the big vendor tool manifests in a server-side registry and reveals only what the agent asks for.
- **Full-stack Ballerina.** Agent, proxy, mock backends, and the 7-service mesh are all Ballerina 2201.13.3 — one language, one build.

◆ The best-practice claim

Cross-system correlation is a single-context reasoning task. Keep it in ONE agent — federate the tools, don't fragment the reasoning.

1

MCP endpoint the agent sees

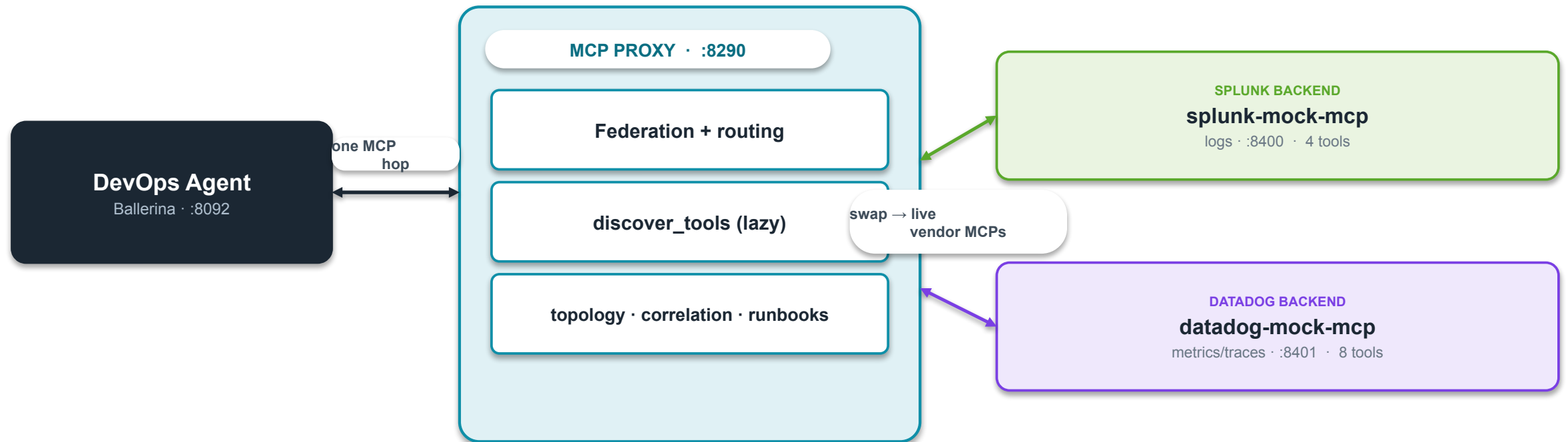
12

topology tools

1

language

Architecture



Why a proxy?

Small agent context

The vendor MCPs expose 50+ tools each. Injecting all of them would blow the context window. The proxy keeps them server-side.

Mock $\rightleftharpoons$ live isolation

One env var on the proxy switches mock to SaaS. The agent is untouched, so you can stage a rollout safely.

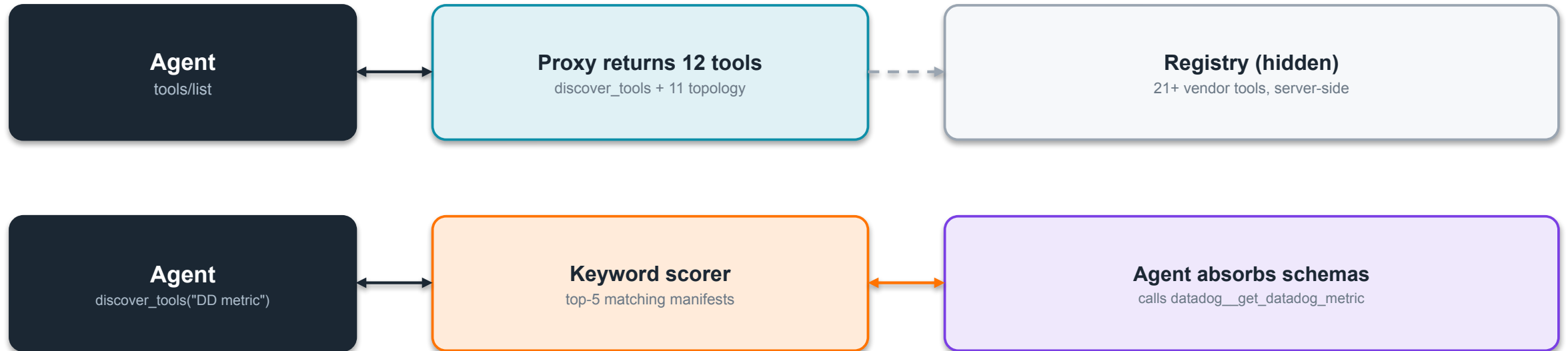
Single policy chokepoint

Routing, auth (via a gateway), and result hygiene get exactly one home instead of being smeared across the agent.

Fills the vendor gap

It's Ballerina, so it can also ACT — hit chaos endpoints, run runbooks — and it owns the catalog the vendors don't have.

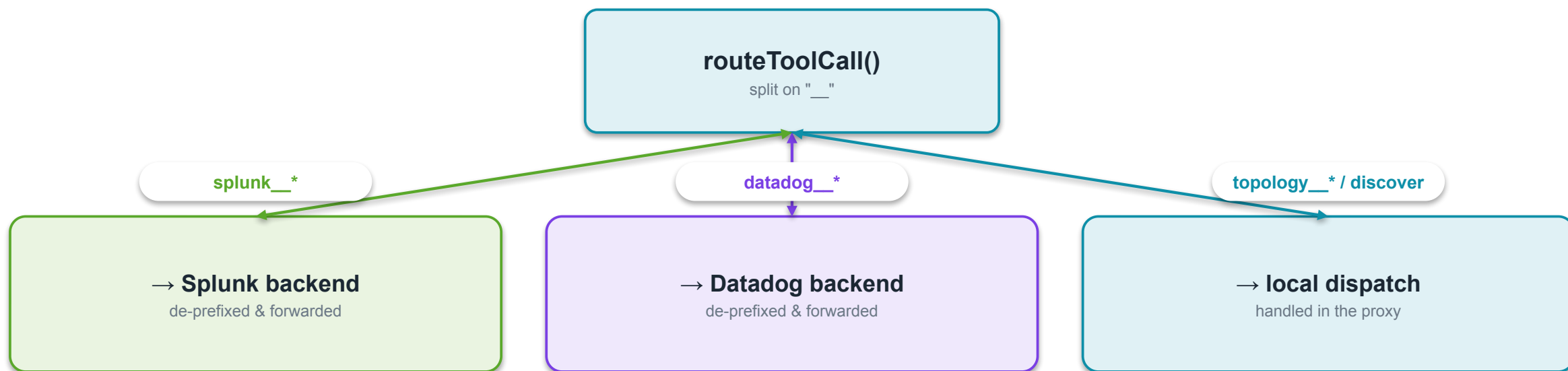
Lazy tool loading — the low-context pattern



♦ POC honesty

Today the router is a keyword scorer — accurate enough at ~21 tools. Production swaps in a pgvector + embedding semantic router when the live MCPs bring 50+ tools each.

Prefix routing keeps it simple



What the agent can reach: **11 topology__** (lookup / dependencies / health / correlate_trace / deploys / incidents / runbooks / audit) + **4 splunk__** + **8 datadog__**

The investigation loop in action



Stack, tools & runbooks

Layer	Choice
Language / runtime	Ballerina Swan Lake 2201.13.3
MCP transport	Streamable HTTP · JSON-RPC 2.0
Telemetry	ballerinax/jaeger (OTLP) + prometheus
LLM loop	native tool-use, no SDK · maxTurns 30
Default model	Ollama qwen (local) / claude-sonnet-4-6

Runbooks — the fixed allowlist

disable-chaos

LIVE — POST /chaos/reset. The demo's recovery lever.

restart-service

stub — kubectl rollout restart

clear-cache

stub — Redis FLUSHDB on inventory

freeze-deploys

stub — sets an in-memory flag

Every execution appends to an in-memory audit log. No run_cli, ever.

Strengths & honest limits

✓ Strengths

- **Small, stable context** no matter how many tools the real MCPs expose.
- **Correlation in one context** logs + spans + topology join in one working set.
- **Mock ⇌ live is config** one env var, proxy-side.
- **Single policy chokepoint** routing, auth, hygiene in one place.
- **Bounded remediation** typed runbook allowlist + approval gate.
- **Clean migration path** wrap with A2A at org boundaries later.

⚠ Known limits (POC)

- **Keyword router** not vector-based; fine at ~21 tools, upgrade for 50+.
- **No result hygiene** live payloads need truncation/neutralization.
- **No endpoint auth** trusted local net; prod defers to the WSO2 gateway.
- **Static catalog** in-code map today; production reads a CMDB.
- **Stub runbooks** only disable-chaos is wired to a real action.
- **SSE buffering risk** a buffering gateway breaks streaming run_runbook.

SECTION

LangChain + A2A

An orchestrator that delegates to Datadog & Splunk specialist agents over the A2A protocol.

The core idea

- **Three agents.** An orchestrator (DevOpsOverSightAgent) delegates to two specialists — a DataDogAgent and a SplunkAgent — over the A2A protocol.
- **Low-context by decomposition.** Each specialist eagerly loads only its own small toolset (8 Datadog / 4 Splunk). The vendor manifests never enter the orchestrator's context.
- **No proxy, no discover_tools.** The proxy dissolves: federation becomes the A2A boundary; correlation & runbooks stay in-process in the orchestrator.
- **Familiar ecosystem.** Python 3.12, LangChain 1.x on the LangGraph runtime — the stack most teams already know.

◆ Same principle, other shape

A2A is used ONLY at the platform-team boundary — not to split correlation. The fusion still happens in one orchestrator context.

3

agents

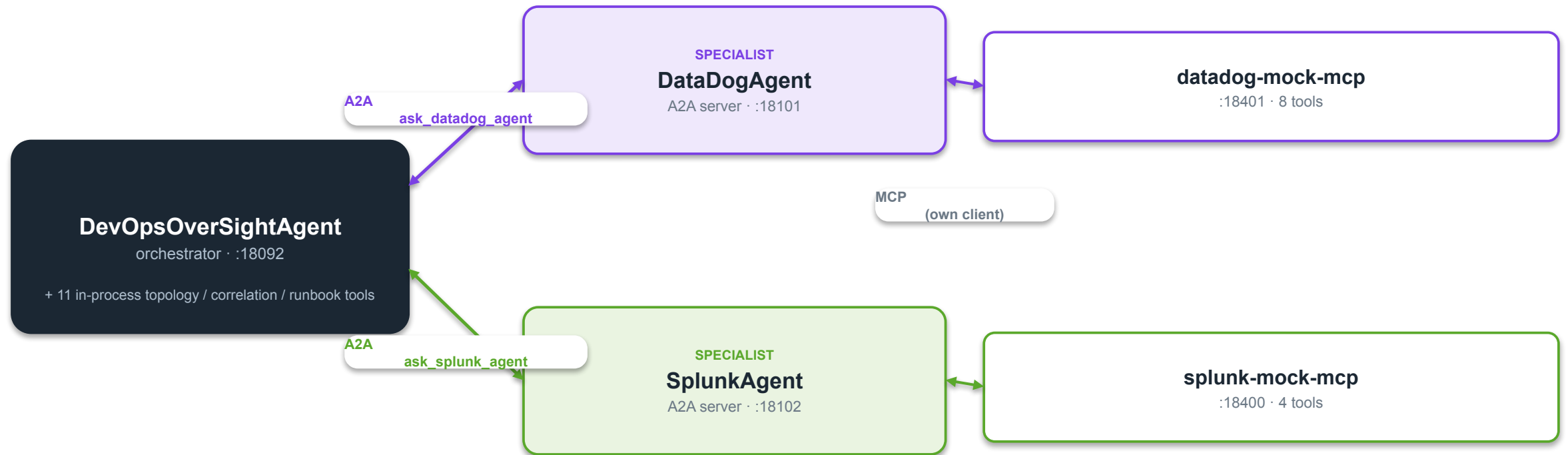
8+4

specialist tools

Python 3.12

LangChain 1.x · LangGraph

Architecture



The A2A protocol

- **Agent cards.** Each specialist publishes `/.well-known/agent-card.json` declaring its skill (`datadog_evidence`, `splunk_log_search`) and capabilities.
- **JSON-RPC delegation.** The orchestrator resolves the card, then sends a message and streams the reply. One-way: orchestrator → specialist.
- **Request/response, not long-running tasks.** Each call is a single round-trip — verified against the SDK, no Task lifecycle.
- **The output contract is load-bearing.** Evidence crosses as prose, so specialists MUST return `trace_ids` verbatim, metric values, timestamps. A vague reply starves the orchestrator.

The delegation call

orchestrator LLM

- `ask_datadog_agent(...)`
- `A2ACardResolver.get_card()`
- `SendMessageRequest` (JSON-RPC)
- specialist ReAct loop → MCP

◆ Version pin matters

`a2a-sdk ≥1.1, <2` is protobuf-based. The older 0.2.x tutorials do NOT apply — build against the installed SDK.

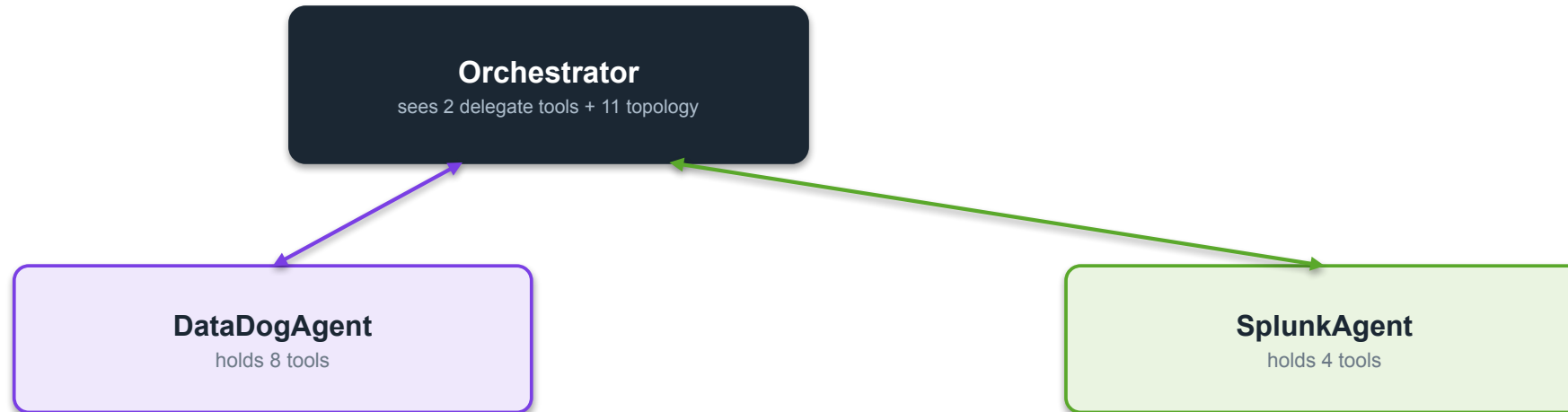
Low-context by decomposition

The problem (shared)

Real vendor MCPs expose 50+ tools each. Load them all into one agent and the context window is gone.

S2's answer

Give each platform its own agent. The orchestrator only ever sees 2 delegate tools, never the vendor manifests.

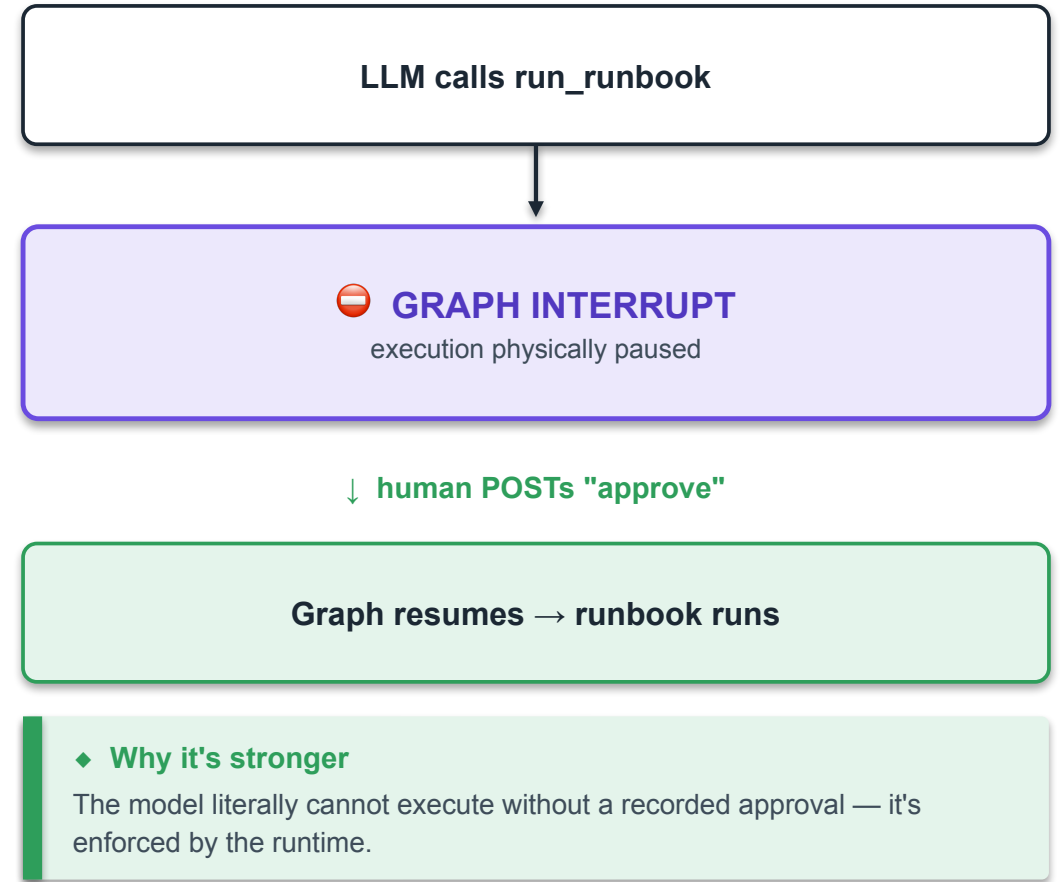


♦ Trade

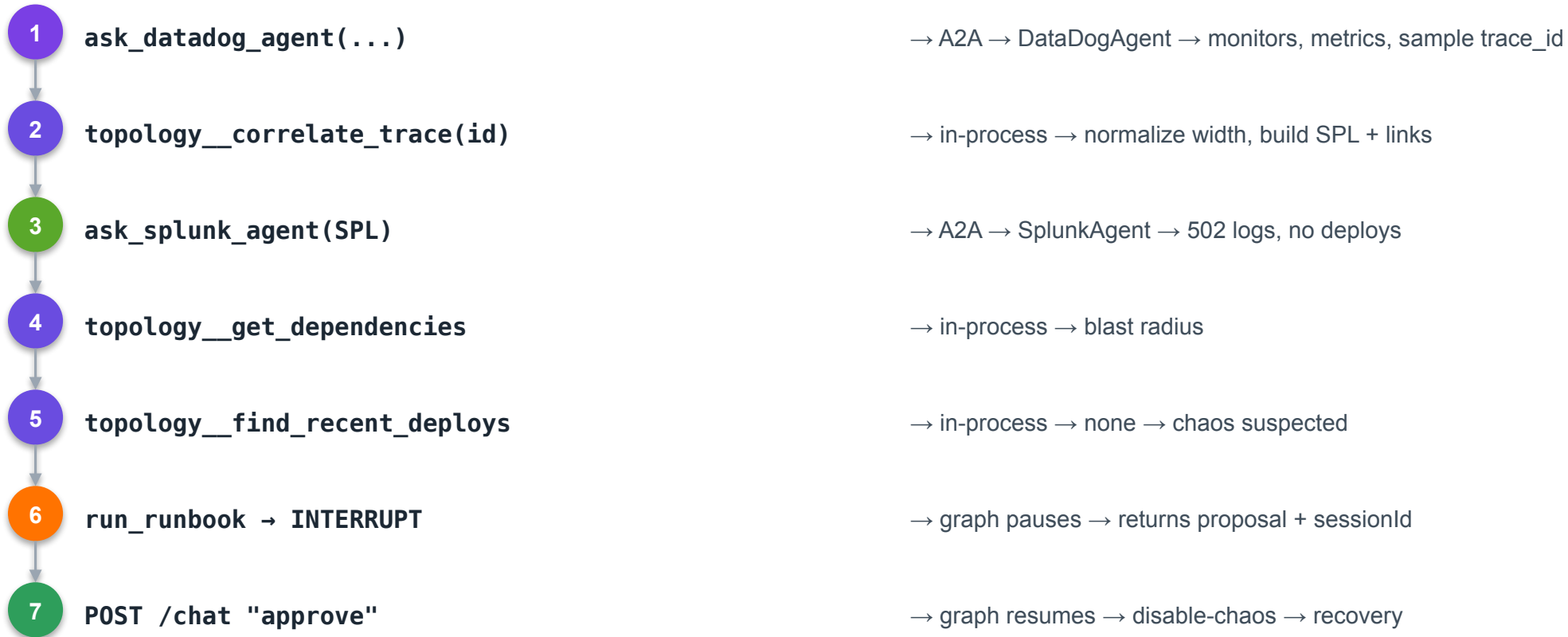
Cleaner separation & familiar code — at the cost of another network hop and 3× the model non-determinism.

The hard approval gate

- **Code-level, not prompt-level.** A LangGraph `HumanInTheLoopMiddleware` physically interrupts the graph before `topology__run_runbook` executes.
- **The graph pauses.** `/investigate` returns the proposal + a `sessionId`. State is checkpointed by `InMemorySaver` keyed to that session.
- **Human resumes it.** Operator POSTs "approve" to `/chat`; the graph resumes with a `Command(resume=...)` and only then runs the runbook.
- **Fail-safe by default.** Any non-approval reply is treated as rejection. The runbook does not run.



The investigation loop with A2A

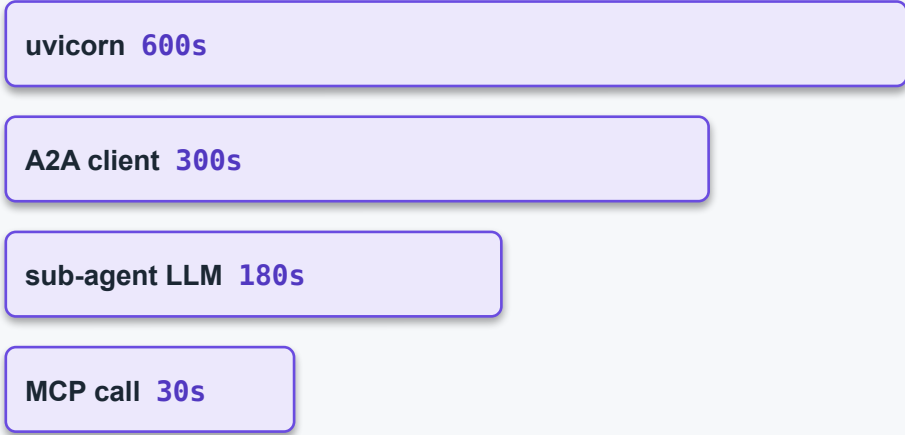


A2A delegation · in-process tool · gate

Stack & the Timeout Chain

Layer	Choice
Language / runtime	Python 3.12 · single uv workspace
Agent framework	LangChain 1.x on LangGraph 1.x (create_agent)
A2A	a2a-sdk ≥1.1, <2 (protobuf)
MCP	official mcp SDK · langchain-mcp-adapters
Gate / state	HumanInTheLoopMiddleware + InMemorySaver

The Timeout Chain (must stay ordered)



Largest-outermost. assert_timeout_chain() fails fast at startup if misordered.

Strengths & honest limits

✓ Strengths

- **Familiar ecosystem** Python + LangChain; low ramp for most teams.
- **Hard, code-level gate** graph interrupt, not a prompt instruction.
- **Structural low-context** specialists cap tools by construction.
- **Clean protocol separation** A2A at the boundary, MCP for tools, in-process fusion.
- **Same model flexibility** 4 providers, config-swappable.
- **Full self-observability** one trace spans user → A2A → specialist → MCP.

⚠ Known limits (POC)

- **3× model non-determinism** three agents must each emit clean tool calls.
- **Prose output contract** vague specialist replies starve correlation.
- **Extra network hop** A2A adds latency vs an in-process call.
- **Timeout Chain to maintain** four nested timeouts, order-sensitive.
- **In-memory state** restart mid-approval drops the pending runbook.
- **SDK version drift** a2a-sdk 1.1 is protobuf; pin carefully.

SECTION

Comparison & Path Forward

An honest scorecard, the 5-minute demo, and what production hardening looks like.

The architectural crux

Both keep correlation in ONE reasoning context. They differ only in how they keep the toolset small.

Solution 1 — federate the tools

- **Low-context via** a proxy with a server-side registry + lazy discover_tools.
- **Correlation locus** one Ballerina agent holds all evidence directly.
- **Boundary** one process federates; nothing is fragmented.

Solution 2 — decompose the agents

- **Low-context via** specialist agents that each own a small toolset.
- **Correlation locus** the orchestrator fuses prose evidence from specialists.
- **Boundary** A2A at the platform-team line — NOT inside correlation.

Side-by-side scorecard

Dimension	Solution 1 · Ballerina + Proxy	Solution 2 · LangChain + A2A
Low-context strategy	Proxy lazy-load (server registry)	Agent decomposition (specialists)
Correlation locus	One agent, direct	Orchestrator fuses prose
Approval gate	Prompt-level instruction	Code-level graph interrupt
Language / stack	Ballerina 2201.13.3	Python 3.12 / LangChain 1.x
Moving parts	Fewer — 1 agent + proxy	More — 3 agents + hop
Model non-determinism	×1 (single agent)	×3 (each specialist)
Ecosystem familiarity	Ballerina — niche	Python/LangChain — broad
Tool-sprawl at 50+ tools	Needs vector router upgrade	Handled by decomposition
WSO2 AMP fit	Native agent runtime	Optional LLM gateway only
Ops surface	Single reasoning process	Distributed; timeout chain

Recommendation — measured, not guessed

Metric — qwen2.5:14b local · 18 runs (3 datasets × 6)	LangChain + A2A	Ballerina + MCP Proxy
Correct-proposal rate	11 / 18 (61%)	5 / 18 (28%)
Reliability by dataset	4/6 · 3/6 · 4/6	3/6 · 2/6 · 0/6
Median time-to-proposal *	~62 s	~78 s
LLM calls / investigation *	~19	~9
Cost per LLM call *	~3 s (small contexts)	~9 s (one big context)

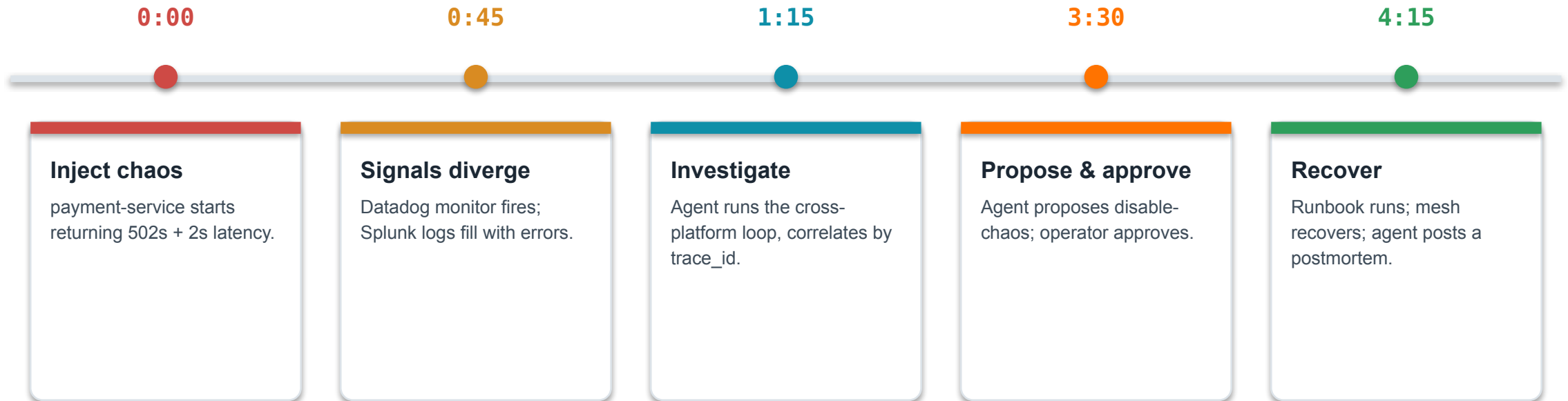
* valid runs only — failed runs bailed out faster but produced no proposal, so timing them would flatter the loser.

What the data says — both my architecture-first predictions were wrong

- **Faster = LangChain** (62 vs 78 s median). I'd predicted Ballerina (fewer calls) — but its ~9 calls are each ~3× heavier (one big, growing context), losing to LangChain's more-but-cheaper calls on small decomposed contexts.
- **More reliable = LangChain, ~2×** (61% vs 28%; Ballerina went 0/6 in one dataset). Opposite my guess too: decomposing into short sub-tasks keeps the 14B on-track better than one long single-agent protocol + discover_tools overhead.
- **Caveat + what still stands** Both are sub-production on a local model → use a cloud model for demo/prod. Qualitative edges unchanged: Ballerina = fewer parts + WSO2-native; LangChain = code-level gate + Python familiarity.

Method: identical qwen2.5:14b-instruct on Ollama; 3 datasets × 6 measured runs each (warm-up discarded per dataset); stacks run sequentially, never concurrent; mock MCP backends; same chaos + payload. Latency via curl, LLM calls via the Ollama access log.

The 5-minute demo — payment-service P1



♦ **Runs offline, resets clean**

No vendor credentials. A token-gated chaos endpoint injects the fault; disable-chaos (or reset) returns to a clean baseline for the next run.

POC → Enterprise: the hardening path

Capability	POC state	Enterprise requirement
Identity & access	Unauthenticated local net	OIDC/SSO + workload identity + per-tool RBAC at the gateway
Secrets	.env variables	Vault / cloud secrets manager; rotation
Audit log	In-memory, session-scoped	Durable, tamper-evident, tied to identity, SIEM-fed
Approval gate	S1 prompt / S2 code-level	Hard code-level interception in both; recorded signal
Service catalog	Static in-code map	CMDB-backed, team-maintained, versioned
Observability backends	Mock MCP servers	Live Splunk (app 7931) + Datadog (Bits AI) MCPs
Trace-ID handling	Verbatim (demo id)	True 64/128-bit normalization on real traffic
Scale	Single node	Agents on K8s; proxy behind LB; managed infra

Cross-cutting gotchas to respect

Trace-ID width (64 vs 128-bit)

The #1 correctness bug. Normalize or the agent sees 'no logs.' Mocks mask it.

NATS async propagation

HTTP trace context auto-propagates; NATS does not. Inject traceparent into the envelope.

Untraced Postgres

Turn on SQL connector tracing or DB latency is invisible to the agent.

SSE buffering at the gateway

A buffering proxy breaks streaming run_runbook progress.

Model non-determinism (Ollama)

Keep maxTurns ≥ 25 ; prefer a cloud model for the live demo.

In-memory state

Audit log & pending approvals are volatile — a restart drops them.

Summary & next steps

What we proved

- **The capability works.** cross-platform diagnosis + approved remediation in a 5-minute, offline, laptop demo.
- **Two credible architectures** — Ballerina + MCP Proxy and LangChain + A2A — each with clear trade-offs.
- **The durable bets hold:** MCP as the contract, single-context correlation, propose-before-act, model portability.
- **A clean path to production** exists for either — a hardening checklist, not a re-architecture.

NEXT STEPS

- 1 Fill in the recommendation with the client's team & governance context.
- 2 Pick one architecture for a hardening spike (auth + live MCPs).
- 3 Wire real Splunk & Datadog MCPs; exercise trace-ID normalization.
- 4 Run the demo live with the client on their own laptop.

Ports & endpoints reference

Solution 1 · Ballerina	Port
DevOps Agent	8092 → 8000
MCP Proxy	8290
splunk-mock-mcp	8400
datadog-mock-mcp	8401
payment chaos endpoint	9196
Prometheus scrape	9797

Solution 2 · LangChain	Port
Orchestrator	18092 → 8000
DataDogAgent (A2A)	18101
SplunkAgent (A2A)	18102
datadog-mock-mcp	18401
splunk-mock-mcp	18400
OTLP collector	14317 / 14318

Shared: demo trace_id **abc123def456789012345678deadbeef** · WSO2 AMP console **:3000** · chaos token **dev-chaos-token**

The LangChain stack 1-prefixes every host port so both stacks run side-by-side for a live A/B on one machine.

Thank you.

Questions, pushback, and better ideas all welcome — this is a team effort.

Scott Bechtel · WSO2 Forward Deployed Engineering